

English
Medium



EDU TERIA

72nd BPSC

Prelims Online Batch

PHYSICS

LECTURE

04

Light – Part - 03



Light – Part - 03

Lens

A lens is a transparent optical medium bounded by two surfaces, at least one of which is curved. It works on the principle of refraction of light.

Types of Lens

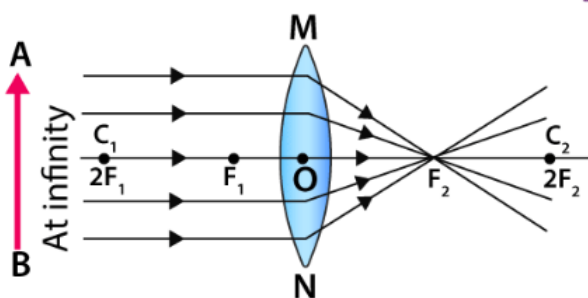
1. Convex Lens
2. Concave Lens

I. Convex lens (Converging)

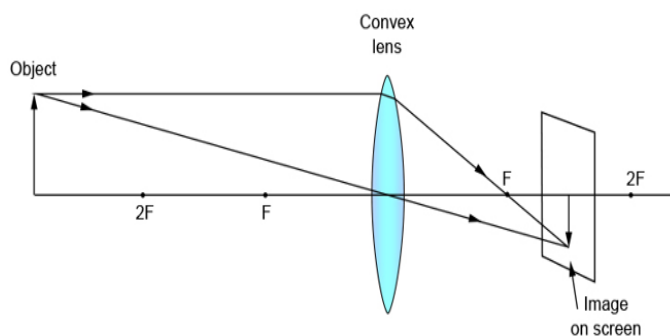
- Forms real & inverted image except when object is inside focus
- Thicker at the center, thinner at edges.
- Used in camera, telescope, microscope, magnifying glass

Image formation in Convex Lens

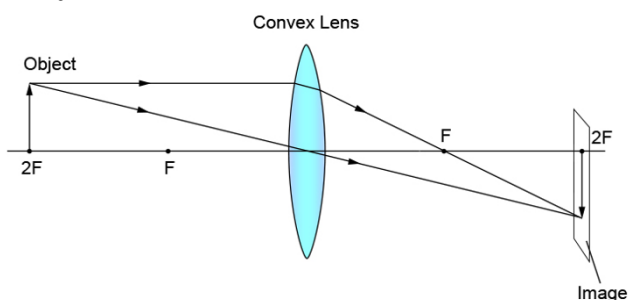
1. Object at ∞



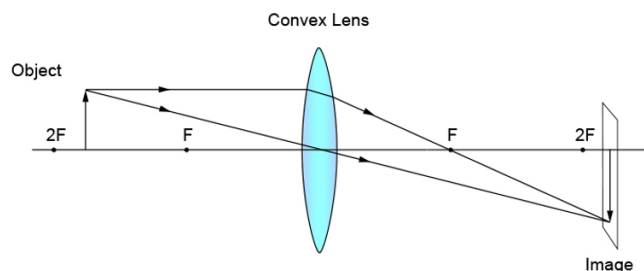
2. Object beyond $2F$



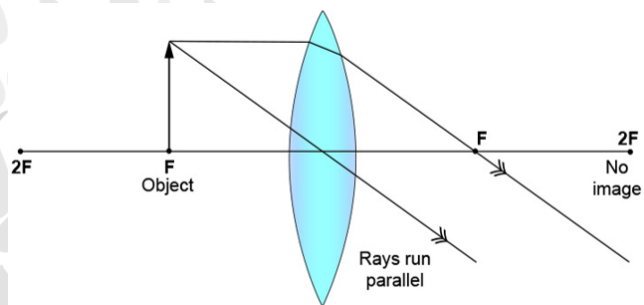
3. Object at $2F$



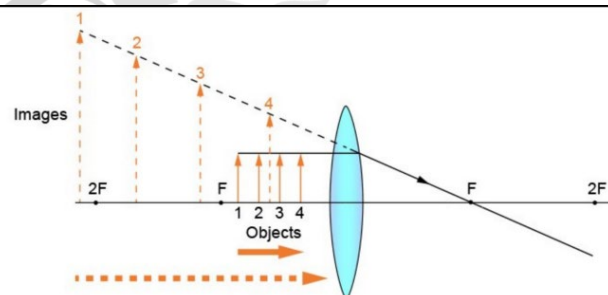
4. Object between F and $2F$



5. Object at F



6. Object between F and optical center (O)



Summary of Image formation of convex lens

Position of Object	Position of Image	Nature of Image	Size of Image
At infinity	At focus (F_2)	Real & inverted	Highly diminished (point image)
Beyond $2F_1$	Between F_2 and $2F_2$	Real & inverted	Diminished
At $2F_1$	At $2F_2$	Real & inverted	Same size
Between F_1 and $2F_1$	Beyond $2F_2$	Real & inverted	Magnified
At F_1	At infinity	Real & inverted	Highly enlarged
Between F_1 and optical centre (O)	On the same side of the lens	Virtual & erect	Magnified

II. Concave lens (Diverging)

- Always forms virtual, erect, diminished images
- Thinner at the center, thicker at edges.
- Used in spectacles for myopia

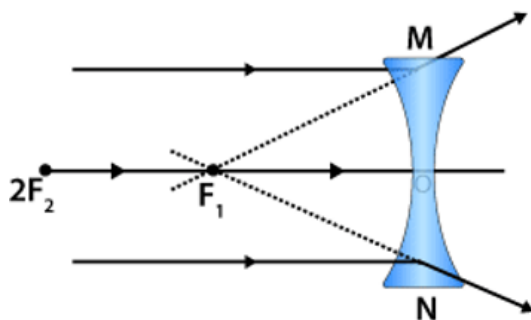


II. Image formation in Concave Lens

1. Object at ∞

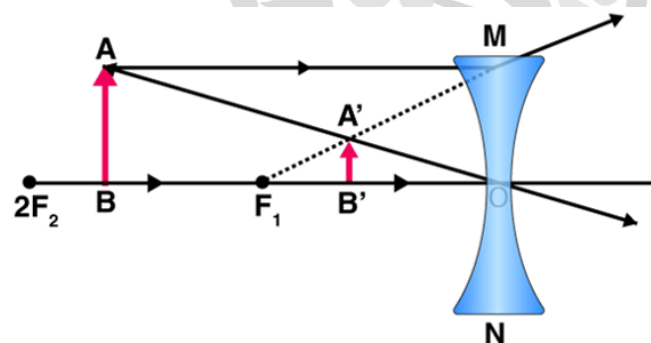
Case – I

- Object – At ∞
- Image – At F_1
- Virtual erect diminished



2. Object between and optical center (O)

Case = II



Summary of Image formation of concave lens

Image formation by Concave Lens			
Object Location	Image Location	Image Nature	Image Size
Infinity	At F_1	Virtual and Erect	Highly diminished, point-sized
Beyond Infinity and Optical Centre	Between Focus (F_1) and Optical center (O)	Virtual and Erect	Diminished

Lens Formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

f= focal

v=Image distance

u=object distance

Magnification

$$m = \frac{h_i}{h_o} = \frac{v}{u}, \quad h_i = \text{Height of image} \quad h_o = \text{Height of object}$$

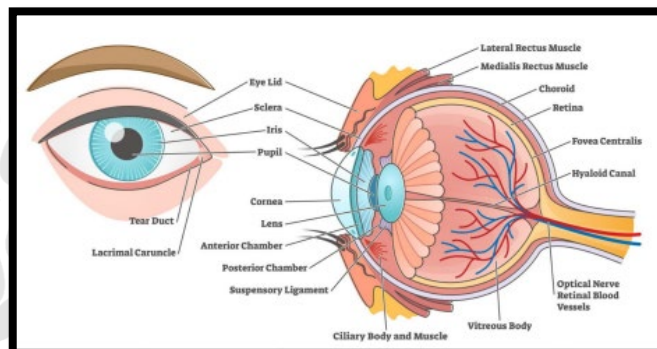
Power of Lens

$$P = \frac{1}{f(\text{in meters})}$$

- Unit : Diopter (D) (-ve)
- Convex lens : +ve power
- Concave lens : -ve power

Human Eye

The human eye is a natural optical instrument that works like a camera. It forms images using refraction of light and allows us to see objects.



Main Components of the Eye

1. Cornea

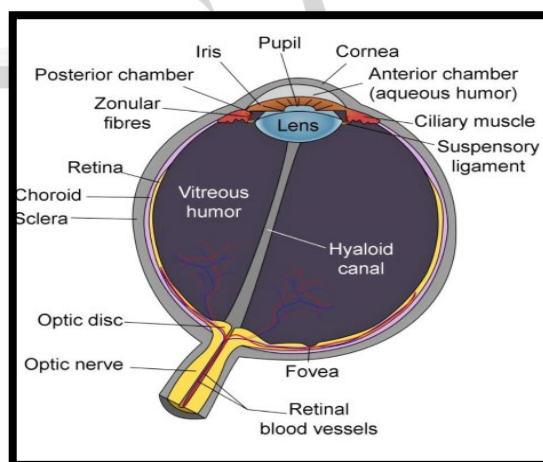
- Transparent, outermost layer
- First surface where light enters
- Provides maximum refraction (bending of light)
- Acts like a fixed convex lens

2. Aqueous Humour

- Clear fluid between cornea and lens
- Maintains intraocular pressure
- Nourishes cornea and lens

3. Iris

- Colored part of the eye
- Controls size of the pupil
- Works like a diaphragm in a camera



**4. Pupil**

- Central opening in the iris
- Regulates amount of light entering the eye
- Expands in dim light, contracts in bright light

5. Eye Lens

- Transparent, flexible convex lens
- Focuses light onto retina
- Can change shape (Accommodation):
- Thick → near objects
- Thin → distant objects

6. Ciliary Muscles

- Surround the lens
- Control its curvature
- Responsible for power of accommodation

7. Vitreous Humour

- Jelly-like substance behind the lens
- Maintains shape of eyeball
- Helps in light transmission

8. Retina

- Light-sensitive inner layer
- Contains photoreceptors:
- Rods → vision in dim light
- Cones → color vision
- Image formed here is real, inverted, and diminished

9. Fovea (Yellow Spot)

- Central part of retina
- Highest concentration of cones
- Provides sharpest vision

10. Blind Spot

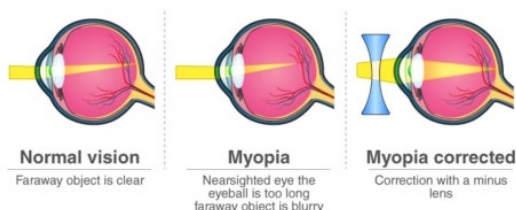
- Point where optic nerve leaves eye
- No photoreceptors → no vision

11. Optic Nerve

- Carries visual signals to brain
- Connects eye to visual cortex

Common Eye Defects**I. Myopia (Nearsightedness)**

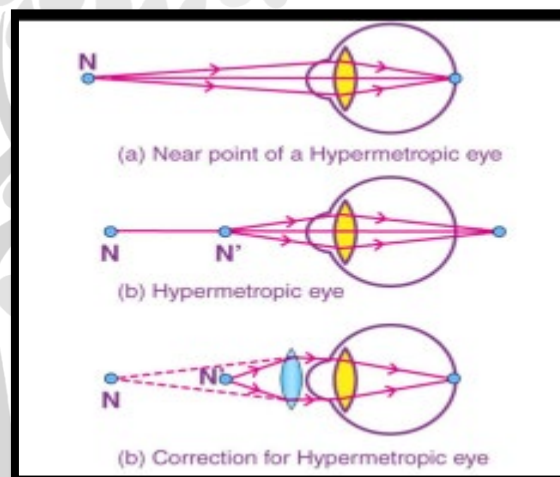
- **Correction:** Concave lens
- **Myopia (Nearsightedness):** Distant objects appear blurry. Example: A student cannot read the whiteboard from the back of the classroom.

**II. Hypermetropia (Farsightedness)**

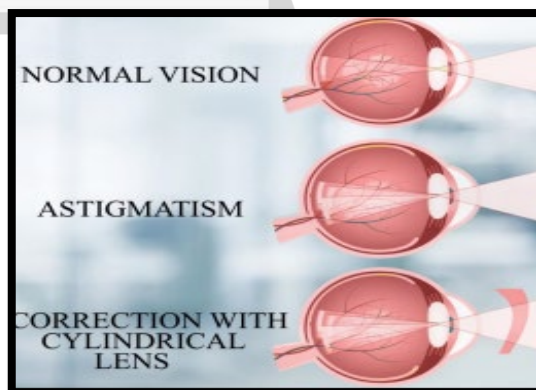
- **Hyperopia (Farsightedness):** Nearby objects appear blurry, while distant objects are clear. Example: A person cannot read a book without holding it at arm's length.
- **Correction:** Convex lens

**III. Presbyopia**

- **Presbyopia:** Age-related inability to focus on close objects, often starting after age 40. **Example:** Needing reading glasses for small print.
- **Correction:** Bifocal lenses

**IV. Astigmatism**

- **Astigmatism:** Blurry or distorted vision at all distances due to an unevenly shaped cornea. **Example:** Seeing lights at night with halos or streaks
- **Correction:** Cylindrical lens



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